

## Lab 4: Text classification with Logistic Regression & SVMs (June 9-12, 2024)

Evaluation: 5% of the total grade

In this lab exercise, you will build text classification models with multiple algorithms using the “20newsgroups” dataset from sklearn. Details of the dataset can be found at the following link:

Data URL: [https://scikit-learn.org/1.4/modules/generated/sklearn.datasets.fetch\\_20newsgroups.html](https://scikit-learn.org/1.4/modules/generated/sklearn.datasets.fetch_20newsgroups.html)

Make sure to follow all the steps outlined in the ML workflow as discussed in the class.

You are tasked to do the following in this lab:

1. Prepare / vectorize the text data to facilitate text classifier development
2. Train and evaluate classification models using logistic regression and support vector machines algorithms.
3. Compare the performance of these models in terms of speed of execution (time), accuracy, precision, recall, F1-scores and confusion-matrix for the test data.
4. Which model/algorithm would you recommend for this application and why?
5. What would you do next to improve the model performance?

**Submit your lab report along with the source code preferably as Jupyter notebook in pdf format by June 20, 2025 @ 11:59.**

### Lab Grading Criteria

| Fraction                   | 0.8 – 1                                                                          | 0.4 – 0.7                                                                            | 0 – 0.3                                                                                 | Max Score |
|----------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------|
| Category                   |                                                                                  |                                                                                      |                                                                                         |           |
| Model development workflow | Followed all steps in the modeling process                                       | Followed most steps in the modeling process                                          | No order in the modeling process                                                        | 1         |
| Tasks and observations     | Performed all the tasks correctly, results properly presented and well explained | Performed most of the tasks correctly, results fairly presented and explained        | Performed some of the tasks correctly, poorly presented results with little explanation | 3         |
| Code                       | Well organized, easy to follow and clean code with meaningful comments           | Somewhat organized, a little hard to follow and fairly clean code with some comments | Not organized, very hard to follow and messy code with little or no comments            | 1         |
| Total                      |                                                                                  |                                                                                      |                                                                                         | 5         |